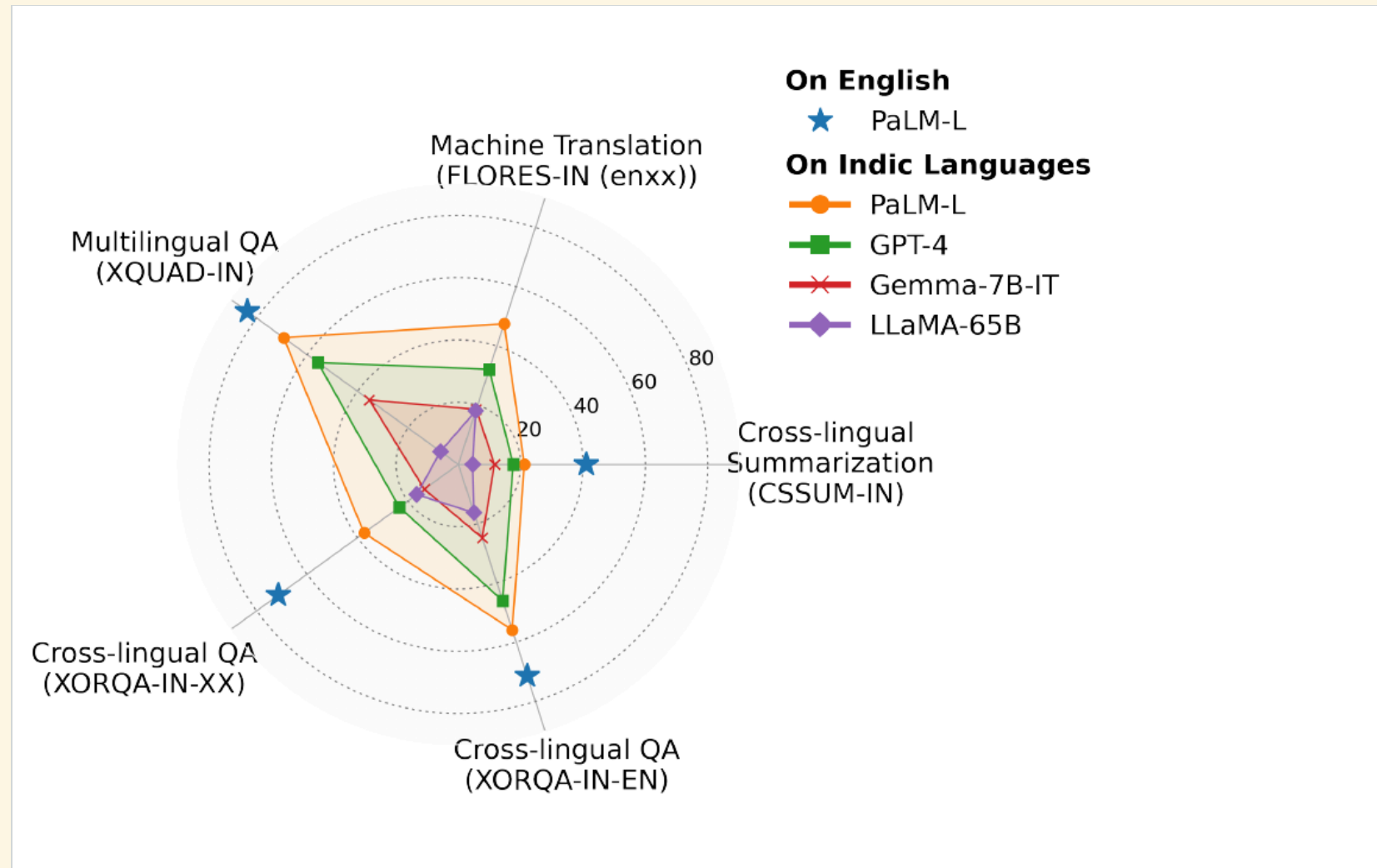


IndicGenBench: Dataset Preparation & Benchmarking

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The gap this benchmark exists to measure



Singh et al. (ACL 2024) built IndicGenBench to make this gap impossible to miss: on every one of five **generation** tasks, even the best model (PaLM-2-L, orange) scores far below its own English performance (blue stars).

Unlike IndicXTREME's classification/NLU focus, this benchmark targets the tasks users actually interact with — **summarizing, translating, and answering questions** in their own language.

Scope: 29 languages, 3 resource tiers

Code	Language	Script	Family	Resource
en	English	Latin	Germanic	high
bn	Bengali	Bengali	Indo-European	higher (Indic relative)
gu	Gujarati	Gujarati	Indo-European	higher (Indic relative)
hi	Hindi	Devanagari	Indo-European	higher (Indic relative)
kn	Kannada	Kannada	Dravidian	higher (Indic relative)
ml	Malayalam	Malayalam	Dravidian	higher (Indic relative)
mr	Marathi	Devanagari	Indo-European	higher (Indic relative)
ta	Tamil	Tamil	Dravidian	higher (Indic relative)
te	Telugu	Telugu	Dravidian	higher (Indic relative)
ur	Urdu	Arabic	Indo-European	higher (Indic relative)
as	Assamese	Bengali	Indo-European	med
bho	Bhojpuri	Devanagari	Indo-European	med
ne	Nepali	Devanagari	Indo-European	med
or	Odia	Odia	Indo-European	med
pa	Punjabi	Gurumukhi	Indo-European	med
ps	Pashto	Arabic	Indo-European	med
sa	Sanskrit	Devanagari	Indo-European	med
awa	Awadhi	Devanagari	Indo-European	low
bgc	Haryanvi	Devanagari	Indo-European	low
bo	Tibetan	Tibetan	Sino-Tibetan	low
brx	Bodo	Devanagari	Sino-Tibetan	low
gbm	Garhwali	Devanagari	Indo-European	low
gom	Konkani	Devanagari	Indo-European	low
hne	Chhattisgarhi	Devanagari	Indo-European	low
hoj	Rajasthani	Devanagari	Indo-European	low
mai	Maithili	Devanagari	Indo-European	low
mni	Manipuri	Meithi	Sino-Tibetan	low
mup	Malvi	Devanagari	Indo-European	low
mwr	Marwari	Devanagari	Indo-European	low
sat	Santali	Ol Chiki	Austroasiatic	low

29 Indic languages, 13 scripts, 4 families — classified into **Higher (9)**, **Medium (7)**, and **Low (13)** resource tiers, based on web-text availability and Joshi et al.'s (2020) taxonomy.

Even the "higher" tier is modest by global standards: Hindi has ~160K Wikipedia articles versus 6.6M+ for English — the paper is explicit that "higher" here means higher *relative to other Indic languages*, not high-resource in any absolute sense.

Five generation tasks, real examples

Task	Language	Input	Output	#Languages (H / M / L)	Dataset Size (Train / Dev / Test)
CROSSSUM-IN (Cross-lingual Summarization)	Hindi	Article: Nasa said a problem with a seismometer provided by the French space agency meant the launch could not go ahead. It is feared the mission could now face a two-year delay. The InSight spacecraft was scheduled to take off between 4-30 March and land on the Red Planet six months later to examine Mars' geology in depth. Nasa said it had decided to call off the launch because the agency was unable to fix a leak affecting the seismometer, which required a vacuum seal to cope with harsh [.....]	अमरीकी अंतरिक्ष एजेंसी नासा अपने अंतरिक्ष यान डिस्कवरी में आई उस समस्या की जांच कर रही है जिसकी वजह से बुधवार को निर्धारित उड़ान स्थगित करनी पड़ी.	9 / 7 / 13	2.9k / 2.9k / 14.5k
FLORES-IN (Machine Translation)	Konkani	She gained critical acclaim during her time in Atlanta and was recognized for innovative urban education.	अटलांटात तिच्या कारकिर्दीत तिका समिक्षकांची तोखणाय मेळळी आनी नाविन्यपूर्ण शारी शिक्षणा खातीर तिका नामनां मेळळीं	9 / 7 / 13	- / 28.9k / 29.3k
XQUAD-IN (Multilingual QA)	Punjabi	Question: ਓਪਰੇਸ਼ਨ ਗਲੇਰੀ ਦਾ ਮਕਸਦ ਕੀ ਸੀ? Context: ਜੰਗ ਤੋਂ ਬਾਅਦ, ਜੁਲਾਈ ਤੋਂ ਨਵੰਬਰ 1954 ਤੱਕ ਓਪਰੇਸ਼ਨ ਗਲੇਰੀ ਚਲਾਇਆ ਗਿਆ, ਤਾਂ ਜੋ ਜੰਗ ਵਿੱਚ ਸ਼ਾਮਲ ਦੇਸ਼ ਆਪਣੇ ਸ਼ਹੀਦਾਂ ਦੀਆਂ ਲਾਸ਼ਾਂ ਨੂੰ ਆਪਣੇ ਦੇਸ਼ ਵਾਪਸ ਲਿਆ ਸਕਣ। ਇਸ ਤਹਿਤ ਯੂ.ਐੱਸ. ਫ਼ੌਜ ਅਤੇ ਯੂ.ਐੱਸ. ਸਮੁੰਦਰੀ ਫ਼ੌਜੀ ਪਲਟਨ ਦੇ 4,167 ਸਿਪਾਹੀ ਅਤੇ 13,528 KPA ਅਤੇ PVA ਦੇ ਸ਼ਹੀਦਾਂ ਦੀਆਂ ਲਾਸ਼ਾਂ ਨੂੰ ਵਾਪਸ ਲਿਆਂਦਾ ਗਿਆ। ਨਾਲ ਹੀ, ਜੰਗ ਦੌਰਾਨ ਸੰਯੁਕਤ ਰਾਸ਼ਟਰ ਦੇ ਕੈਦੀਆਂ ਦੇ ਕੌਪ ਵਿੱਚ ਮਰਨ ਵਾਲੇ 546 ਨਾਗਰਿਕਾਂ ਨੂੰ ਦੱਖਣੀ ਕੋਰੀਆ ਦੀ ਸਰਕਾਰ ਨੂੰ ਸੌਂਪਿਆ ਗਿਆ ਸੀ। [.....]	ਜੰਗ ਵਿੱਚ ਸ਼ਾਮਲ ਦੇਸ਼ ਆਪਣੇ ਸ਼ਹੀਦਾਂ ਦੀਆਂ ਲਾਸ਼ਾਂ ਨੂੰ ਆਪਣੇ ਦੇਸ਼ ਵਾਪਸ ਲਿਆ ਸਕਣ	9 / 3 / -	1.2k / 1.2k / 14.3k
XORQA-IN-XX (Cross-lingual QA)	Telugu	Question: 1989 లో భారత ప్రధాని ఎవరు? Context: Rajiv Ratna Gandhi (; 20 August 1944 – 21 May 1991) was an Indian politician who served as the 6th Prime Minister of India from 1984 to 1989. He took office after the 1984 assassination of his mother, Prime Minister Indira Gandhi, to become the youngest Indian Prime Minister at the age of 40.	రాజీవ్ రత్న గాంధీ	9 / 6 / 13	2.8k / 14k / 15.1k
XORQA-IN-EN (Cross-lingual QA)	Santali	Question: ଭେଲିଆ ଭେଲିଉଆ ୧ ୨୧ ୧୩୦୩୧ ୮୧ ୩୭୭ ୦୮୩୩୦ ୨୧ ୨୮୩ ୧୩୧୩୧୩ ୨୧୩୩ ୧୩୧୩ ୦୩୧୩ ୮୩୧୩ Context: Tibet had been under the administrative rule of the Qing dynasty since 1720. [.....]. In early 1912, the Government of the Republic of China replaced the Qing dynasty as the government of China and the new republic asserted its sovereignty over all the territories of the previous dynasty, which included 22 Chinese provinces, Tibet and Mongolia.	Government of the Republic of China	9 / 6 / 13	2.8k / 14k / 15.1k

Cross-lingual summarization, machine translation, multilingual QA, and two flavors of cross-lingual QA – each with genuine dev/test data collected specifically for this benchmark, plus a small training set released for every task except FLORES-IN.

Why translate existing benchmarks, not build from scratch?

Every IndicGenBench dataset starts from an existing **English**-source benchmark — CrossSum, FLORES-200, XQuAD, XOR-TyDi — and adds professional human translations into Indic languages. The paper gives three explicit reasons:

1. MULTI-WAY PARALLEL DATA

Translating the same source examples into every language lets researchers separate *task knowledge* from *language understanding*, and directly measure cross-lingual generalization.

2. NO CLEAN SOURCE TEXT EXISTS

For many low-resource languages, there's no clean knowledge corpus (e.g. Wikipedia) to draw fresh source material from — translation sidesteps a data-acquisition problem that creating from scratch cannot.

3. INHERIT EXISTING QUALITY CONTROL

By only translating into the target languages, the benchmark leverages all the design and quality-control work that already went into the source benchmarks.

Who did the translating

Annotators were professional data labelers — contractors at the authors' organization and via a vendor — paid competitive rates compliant with applicable labor laws and prevailing market rates in each region.

A 5.7× PAY RANGE ACROSS LANGUAGES

Pay rates varied by language, from **USD 2.80/hour for Pashto** to **USD 15.90/hour for Tibetan** — reflecting local market rates rather than a single global rate, a detail worth noticing when comparing annotation practices across benchmarks.

Creating CrossSum-IN

Source: CrossSum (Bhattacharjee et al., 2023), itself derived from XL-Sum — BBC news articles in one language paired with summaries in another.

1. Sample 700 English article-summary pairs (100 train, 100 dev, 500 test).
2. For the 9 target languages CrossSum *already* covers, resample 100/100/500 from the existing data instead — keeping every language's dataset the same size.
3. For the remaining 20 languages, ask human translators to translate the English summary into the target language.

700 examples × 29 languages = **20,300** examples total — confirmed by direct computation from the paper's own per-language and language-count figures.

A CrossSum-IN example

Task	Language	Input	Output	#Languages (H / M / L)	Dataset Size (Train / Dev / Test)
CROSSSUM-IN (Cross-lingual Summarization)	Hindi	Article: Nasa said a problem with a seismometer provided by the French space agency meant the launch could not go ahead. It is feared the mission could now face a two-year delay. The InSight spacecraft was scheduled to take off between 4-30 March and land on the Red Planet six months later to examine Mars' geology in depth. Nasa said it had decided to call off the launch because the agency was unable to fix a leak affecting the seismometer, which required a vacuum seal to cope with harsh [.....]	अमरीकी अंतरिक्ष एजेंसी नासा अपने अंतरिक्ष यान डिस्कवरी में आई उस समस्या की जाँच कर रही है जिसकी वजह से बुधवार को निर्धारित उड़ान स्थगित करनी पड़ी.	9 / 7 / 13	2.9k / 2.9k / 14.5k

The color-coded spans show the alignment CrossSum was built on: "Nasa," "problem," "launch," "spacecraft," and "scheduled" in the English article each map to their counterpart in the Hindi summary — the model's job is to produce this kind of faithful, compressed summary directly in the target language, never via an intermediate English summary.

Creating FLORES-IN

Source: FLORES-200 (NLLB-Team et al., 2022) — human-annotated, multi-way parallel MT data, already covering 22 of the 29 target languages.

REUSE WHERE POSSIBLE, TRANSLATE WHERE NOT

FLORES-200 splits into dev (997), devtest (1012), and a non-public test (992) set. All 997 dev + 1012 devtest sentences are used, giving **2,009** sentences per language. For the remaining 7 new languages, the same sentences are freshly translated by humans.

$2,009 \times 29 \text{ languages} = \mathbf{58,261} \approx 58.2\text{k}$ examples — matches the paper. FLORES-IN is the only task with no training split.

A FLORES-IN example

Task	Language	Input	Output	#Languages (H / M / L)	Dataset Size (Train / Dev / Test)
FLORES-IN (Machine Translation)	Konkani	She gained critical acclaim during her time in Atlanta and was recognized for innovative urban education.	अटलांटात तिच्या कारकिर्दीत तिका समिक्षकांची तोखणाय मेळ्ळी आनी नाविन्यपूर्ण शारी शिक्षणा खातीर तिका नामनां मेळ्ळीं	9 / 7 / 13	- / 28.9k / 29.3k

Plain sentence-level translation — no context, no question, just source sentence in, target sentence out. This same English sentence has a translation like this one in every one of the 29 languages, which is exactly what makes FLORES-IN *n-way parallel*: every language's output is directly comparable, because they all started from the identical source sentence.

Creating XQuAD-IN

Source: XQuAD (Artetxe et al., 2020), itself derived from SQuAD — an English Wikipedia passage paired with multiple question-answer pairs, where answers are short spans.

1. Reuse XQuAD's existing 240 passages / 1,190 QA pairs (already human-translated into 10 higher-resource languages, Hindi being the only Indic one) as the test set.
2. Additionally select 20 passages / 100 QA pairs from SQuAD's own training set, *and* another 20 passages / 100 QA pairs from its dev set, to build a train and dev split.
3. Collect professional human translations of all 280 passages / 1,390 QA pairs into 12 Indic languages.

$280 \times 12 = \mathbf{3,360} \approx 3.3\text{k}$ passages; $1,390 \times 12 = \mathbf{16,680} \approx 16.6\text{k}$ QA pairs — matches the paper exactly.

An XQuAD-IN example: "multilingual" means everything matches

Task	Language	Input	Output	#Languages (H / M / L)	Dataset Size (Train / Dev / Test)
XQUAD-IN (Multilingual QA)	Punjabi	Question: ਓਪਰੇਸ਼ਨ ਗਲੋਰੀ ਦਾ ਮਕਸਦ ਕੀ ਸੀ? Context: ਜੰਗ ਤੋਂ ਬਾਅਦ, ਜੁਲਾਈ ਤੋਂ ਨਵੰਬਰ 1954 ਤੱਕ ਓਪਰੇਸ਼ਨ ਗਲੋਰੀ ਚਲਾਇਆ ਗਿਆ, ਤਾਂ ਜੋ ਜੰਗ ਵਿੱਚ ਸ਼ਾਮਲ ਦੇਸ਼ ਆਪਣੇ ਸ਼ਹੀਦਾਂ ਦੀਆਂ ਲਾਸ਼ਾਂ ਨੂੰ ਆਪਣੇ ਦੇਸ਼ ਵਾਪਸ ਲਿਆ ਸਕਣ। ਇਸ ਤਹਿਤ ਯੂ.ਐੱਸ. ਫ਼ੌਜ ਅਤੇ ਯੂ.ਐੱਸ. ਸਮੁੰਦਰੀ ਫ਼ੌਜੀ ਪਲਟਨ ਦੇ 4,167 ਸਿਪਾਹੀ ਅਤੇ 13,528 KPA ਅਤੇ PVA ਦੇ ਸ਼ਹੀਦਾਂ ਦੀਆਂ ਲਾਸ਼ਾਂ ਨੂੰ ਵਾਪਸ ਲਿਆਂਦਾ ਗਿਆ। ਨਾਲ ਹੀ, ਜੰਗ ਦੌਰਾਨ ਸੰਯੁਕਤ ਰਾਸ਼ਟਰ ਦੇ ਕੈਦੀਆਂ ਦੇ ਕੈਂਪ ਵਿੱਚ ਮਰਨ ਵਾਲੇ 546 ਨਾਗਰਿਕਾਂ ਨੂੰ ਦੱਖਣੀ ਕੋਰੀਆ ਦੀ ਸਰਕਾਰ ਨੂੰ ਸੌਂਪਿਆ ਗਿਆ ਸੀ। [.....]	ਜੰਗ ਵਿੱਚ ਸ਼ਾਮਲ ਦੇਸ਼ ਆਪਣੇ ਸ਼ਹੀਦਾਂ ਦੀਆਂ ਲਾਸ਼ਾਂ ਨੂੰ ਆਪਣੇ ਦੇਸ਼ ਵਾਪਸ ਲਿਆ ਸਕਣ	9 / 3 / -	1.2k / 1.2k / 14.3k

Question, context passage, and answer are all in **Punjabi** — the same language throughout. That's the defining feature of "multilingual" reading comprehension here: the model reads and answers entirely within one language at a time, just repeated across 12 languages. Contrast this with the next task.

Creating XorQA-IN: a two-stage translation

Source: XOR-TyDi (Asai et al., 2021) — a non-English question paired with an **English** evidence passage and answer, designed for cross-lingual retrieval. It covers only two Indic languages: Bengali (302 examples) and Telugu (237).

STAGE 1 — NORMALIZE THROUGH ENGLISH

Translators first convert the original Bengali/Telugu question into **English**.

STAGE 2 — FAN OUT TO EVERY TARGET LANGUAGE

That English question is then translated into *all* target languages, and separately, the English answer is translated into each target language too.

Routing every language through a common English pivot — rather than translating pairwise between 28 languages — is what makes this scale to a benchmark this size at all.

Two tasks from one translation effort

The Stage-2 translations (question and answer, both now available in every target language) are recombined into two distinct evaluation tasks:

TASK	QUESTION	PASSAGE	ANSWER	TESTS
XorQA-IN-EN	target language	English	English	cross-lingual retrieval, matching the original XOR-TyDi setup
XorQA-IN-XX	target language	English	target language	generating the answer in the <i>same</i> language as the question

Same underlying data, two different generation demands on the model — one English-answer task inherited from the source dataset, one same-language-answer task that's new.

An XorQA-IN example: "cross-lingual" means the passage doesn't match

Task	Language	Input	Output	#Languages (H / M / L)	Dataset Size (Train / Dev / Test)
XORQA-IN-XX (Cross-lingual QA)	Telugu	Question: 1989 లో భారత ప్రధాని ఎవరు?			
		Context: Rajiv Ratna Gandhi (; 20 August 1944 – 21 May 1991) was an Indian politician who served as the 6th Prime Minister of India from 1984 to 1989. He took office after the 1984 assassination of his mother, Prime Minister Indira Gandhi, to become the youngest Indian Prime Minister at the age of 40.	రాజీవ్ రత్న గాంధీ	9 / 6 / 13	2.8k / 14k / 15.1k

Task	Language	Input	Output	#Languages (H / M / L)	Dataset Size (Train / Dev / Test)
XORQA-IN-EN (Cross-lingual QA)	Santali	Question: ସଞ୍ଚିତ କାହାଣୀଟିର ନାମ କ'ଣ ଓ ଏହାର ମୁଖ୍ୟ ଚରିତ୍ର କିଏ ? Context: Tibet had been under the administrative rule of the Qing dynasty since 1720. [.....]. In early 1912, the Government of the Republic of China replaced the Qing dynasty as the government of China and the new republic asserted its sovereignty over all the territories of the previous dynasty, which included 22 Chinese provinces, Tibet and Mongolia.	Government of the Republic of China	9 / 6 / 13	2.8k / 14k / 15.1k

In both variants the question is asked in the target language but the evidence passage stays in **English** – the model must cross languages to find the answer at all. Only the output language differs: XorQA-IN-XX answers back in Telugu, XorQA-IN-EN answers in English.

Not every language is in every task

Code	Language	Coverage in INDICGENBENCH				
		CROSSSUM-IN	FLORES-IN	XQUAD-IN	XORQA-IN-XX	XORQA-IN-EN
bn	Bengali	✓	✓	✓	✓	✓
gu	Gujarati	✓	✓	✓	✓	✓
hi	Hindi	✓	✓	✓	✓	✓
kn	Kannada	✓	✓	✓	✓	✓
ml	Malayalam	✓	✓	✓	✓	✓
mr	Marathi	✓	✓	✓	✓	✓
ta	Tamil	✓	✓	✓	✓	✓
te	Telugu	✓	✓	✓	✓	✓
ur	Urdu	✓	✓	✓	✓	✓
as	Assamese	✓	✓	✓	✓	✓
bho	Bhojpuri	✓	✓		✓	✓
ne	Nepali	✓	✓			
or	Odia	✓	✓	✓	✓	✓
pa	Punjabi	✓	✓	✓	✓	✓
ps	Pashto	✓	✓		✓	✓
sa	Sanskrit	✓	✓		✓	✓
awa	Awadhi	✓	✓		✓	✓
bgc	Haryanvi	✓	✓		✓	✓
bo	Tibetan	✓	✓		✓	✓
brx	Bodo	✓	✓		✓	✓
gbm	Garhwali	✓	✓		✓	✓
gom	Konkani	✓	✓		✓	✓
hne	Chhattisgarhi	✓	✓		✓	✓
hoj	Rajasthani	✓	✓		✓	✓
mai	Maithili	✓	✓		✓	✓
mni	Manipuri	✓	✓		✓	✓
mup	Malvi	✓	✓		✓	✓
mwr	Marwari	✓	✓		✓	✓
sat	Santali	✓	✓		✓	✓

CrossSum-IN and FLORES-IN reach all 29 languages. XQuAD-IN — constrained by which languages already had XQuAD translations plus the 3 medium-resource languages selected for extension — covers only 11. XorQA-IN's two tasks reach 28 (Nepali was not translated).

Choosing evaluation metrics that work for low-resource text

CHRF — SUMMARIZATION & TRANSLATION

Character-level F-score (Popović, 2015), used for CrossSum-IN and FLORES-IN. Token-level metrics like ROUGE and BLEU are unreliable for low-resource languages (Bapna et al., 2022) — tokenization itself is often the weak link, so scoring on characters instead sidesteps it.

TOKEN-F1 — QUESTION ANSWERING

SQuAD-style token-overlap F1, used for XQuAD-IN and both XorQA-IN tasks — kept consistent with the rest of the QA literature these benchmarks are drawn from.

Does the benchmark distinguish good models? Yes.

Model (LLM)	CROSSSUM-IN	FLORES-IN	XQUAD-IN	XORQA-IN-XX	XORQA-IN-EN
Eval. Metric	ChrF	ChrF	Token-F1	Token-F1	Token-F1
	(enxx / xxen)				
Performance in <i>English</i>					
GPT-4	30.3	– / –	64.8	–	37.9
PaLM-2-L	41.1	– / –	83.7	–	71.4
Average Performance on INDICGENBENCH					
LLaMA-7B	3.7	11.5 / 21.6	3.8	7.4	10.4
LLaMA-13B	4.1	13.3 / 24.1	4.5	10.4	12.1
LLaMA-65B	4.6	18.1 / 32.7	7.1	16.5	16.3
BLOOM-7B	3.8	18.3 / 31.2	13.8	7.9	23.6
BLOOMZ-7B	1.2	40.8 / 48.4	53.7	7.0	49.0
Gemma-7B-PT	0.0	32.1 / 50.4	0.5	11.7	23.8
Gemma-7B-IT	11.6	18.6 / 29.2	35.3	13.5	24.8
GPT-3.5	16.3	29.2 / 47.7	33.2	21.6	35.5
GPT-4	17.6	32.1 / 54.5	55.7	23.4	46.0
PaLM-2-XXS	7.2	24.0 / 43.4	34.6	13.5	36.8
PaLM-2-XS	15.5	40.7 / 58.3	62.2	29.5	47.8
PaLM-2-S	18.5	43.5 / 61.6	66.7	31.6	57.4
PaLM-2-L	21.2	47.5 / 65.1	69.3	37.4	55.9

On FLORES-IN, we report performance in both directions—the source to target language (enxx to xxen) and the target to source language (xxen to enxx).

3.1 Comparison on INDICGENBENCH

In Table 2 we evaluate the performance of Gemma, GPT and PaLM-2 on the tasks of INDICGENBENCH in the English setting. Numbers are in the evaluation data. We see that PaLM-2-L performs the best across the board, except for the English performance where PaLM-2-S performs better.

We see across the board that open source LLMs underperform compared to the largest LLaMA-65B. PaLM-2-S performs the smallest P

Larger models within a family consistently do better. PaLM-2-L wins on 4 of 5 tasks; open-source LLaMA lags badly (even LLaMA-65B underperforms the smallest PaLM-2-XXS); and BLOOMZ — instruction-tuned, broad multilingual pretraining — wins 3 of 5 tasks outright. Every model, even the best, trails its own English performance by 20+ points.

The drop-off by resource tier — and an asymmetry

Model	CROSSSUM-IN			FLORES-IN (enxx / xxen)			XQUAD-IN		XORQA-IN-XX			XORQA-IN-EN		
	High	Medium	Low	High	Medium	Low	High	Medium	High	Medium	Low	High	Medium	Low
LLaMA-65B	4.4	4.6	4.7	18.2 / 31.5	15.4 / 30.0	19.5 / 35.0	8.8	1.9	17.7	13.5	17.1	16.4	14.0	17.3
Gemma-7B-IT	13.9	11.5	10.0	17.6 / 33.7	15.0 / 26.1	21.3 / 27.7	38.8	24.8	18.9	8.3	12.2	29.5	23.9	21.9
BLOOMZ-7B	1.5	1.7	0.6	67.7 / 59.1	39.4 / 50.2	22.9 / 40.0	55.5	48.1	10.8	2.8	6.2	64.7	45.8	39.5
GPT-4	19.4	17.9	16.3	36.2 / 59.6	30.7 / 55.2	29.9 / 50.5	56.1	54.6	25.8	21.6	22.6	49.4	50.0	41.8
PaLM-2-L	25.2	23.1	17.5	56.9 / 68.2	45.9 / 65.6	41.9 / 62.6	72.5	59.8	41.9	36.7	34.6	57.3	57.9	53.9

Table 2: One-shot performance across language categories based on resource tier, defined in Section 2. For all

Performance drops from High to Medium to Low resource languages across every task. But look at FLORES-IN's two directions for PaLM-2-L: English→target drops sharply (56.9→41.9), while target→English barely moves (68.2→62.6). **Current LLMs are far better at understanding low-resource languages than at generating them.**

More in-context examples help — up to a point

Model (LLM)	FLORES-IN			XORQA-IN-XX			
	0	1	5	0	1	2	3
LLaMA-7B	8.0	11.5	11.4	5.0	7.4	9.0	9.2
LLaMA-13B	8.6	13.3	13.4	6.3	10.4	12.2	13.1
LLaMA-65B	14.0	18.1	18.3	12.3	16.5	18.7	19.4
PaLM-2-XXS	0.8	24.0	26.9	8.9	13.5	15.8	17.5
PaLM-2-XS	20.1	40.7	42.3	21.4	29.5	32.2	33.2
PaLM-2-S	24.9	43.5	45.2	22.7	31.6	33.4	35.4
PaLM-2-L	31.1	47.5	49.3	31.9	37.4	39.7	41.1

Table 4: Performance by varying number of in-

Every model family improves monotonically as more in-context examples are added — the biggest jump is almost always from 0 to 1 shot (e.g. PaLM-2-XXS on FLORES-IN: 0.8 → 24.0).

This sets up the next question: how many shots actually *fit*, once tokenizer fertility eats into the context budget?

Hindi exemplars beat English exemplars

Model (1-Shot Lang)	CROSSSUM-IN			XQUAD-IN		XORQA-IN-XX			XORQA-IN-EN		
	Higher	Medium	Low	Higher	Medium	Higher	Medium	Low	Higher	Medium	Low
PaLM-2-XXS (En)	0.3	0.1	0.3	<u>38.5</u>	<u>31.9</u>	14.0	5.4	7.3	40.3	35.0	30.8
PaLM-2-XXS (Hi)	<u>1.3</u>	<u>2.1</u>	<u>3.7</u>	39.8	33.3	<u>17.6</u>	<u>8.5</u>	<u>10.5</u>	45.5	39.4	31.9
PaLM-2-XXS (Lang)	7.7	7.6	6.7	37.2	26.8	17.7	8.8	12.8	<u>43.6</u>	<u>38.3</u>	<u>31.5</u>
PaLM-2-XS (En)	0.3	0.2	0.5	64.3	<u>62.2</u>	30.6	23.9	20.8	35.9	32.1	27.2
PaLM-2-XS (Hi)	<u>3.5</u>	<u>5.5</u>	<u>9.9</u>	65.4	63.5	<u>33.2</u>	<u>25.8</u>	<u>22.7</u>	<u>49.3</u>	<u>46.8</u>	<u>40.7</u>
PaLM-2-XS (Lang)	18.4	16.4	13.0	<u>65.1</u>	53.3	35.8	27.6	26.1	53.3	51.5	42.2
PaLM-2-S (En)	0.4	0.2	0.5	67.4	<u>66.8</u>	27.5	19.9	19.9	48.6	47.1	40.8
PaLM-2-S (Hi)	<u>4.4</u>	<u>6.9</u>	<u>13.2</u>	<u>68.5</u>	67.5	<u>34.2</u>	<u>27.0</u>	<u>24.9</u>	<u>58.3</u>	<u>57.0</u>	<u>49.0</u>
PaLM-2-S (Lang)	22.4	19.8	15.1	69.9	57.3	36.6	30.3	28.6	60.1	61.4	53.6
PaLM-2-L (En)	0.4	0.2	0.6	71.7	<u>69.8</u>	37.7	33.2	29.7	28.7	27.5	26.2
PaLM-2-L (Hi)	<u>4.7</u>	<u>7.0</u>	<u>13.8</u>	72.6	71.0	<u>39.7</u>	<u>34.6</u>	<u>31.2</u>	<u>45.5</u>	<u>44.8</u>	<u>41.5</u>
PaLM-2-L (Lang)	25.2	23.1	17.5	<u>72.5</u>	59.8	41.9	36.7	34.6	57.3	57.9	53.9

Prompting order is consistent everywhere: **test language > Hindi > English** — exactly the order of linguistic relatedness to the test language. For smaller models, Hindi exemplars come extremely close to (sometimes beat) exemplars in the test language itself.

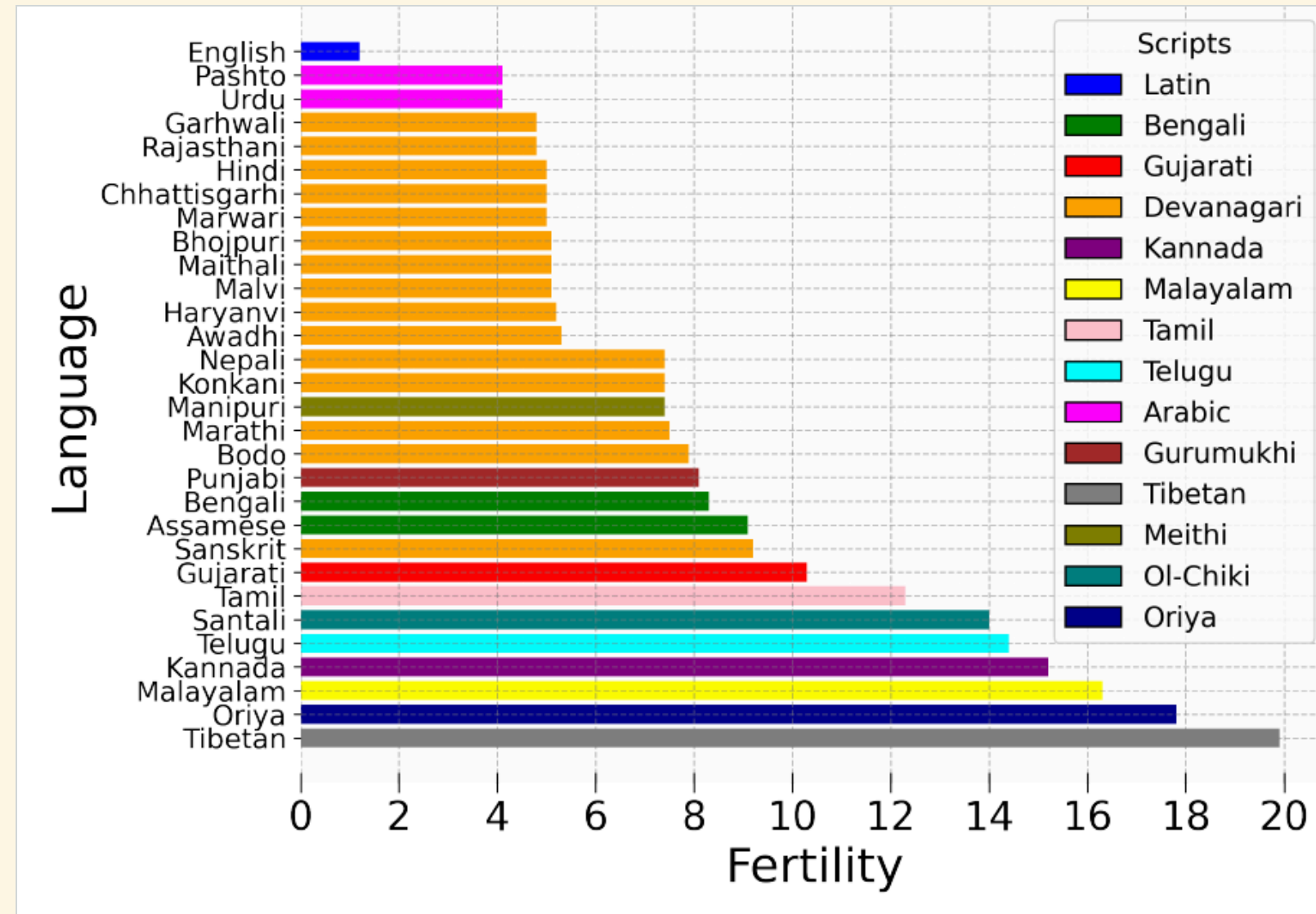
Fine-tuning vs. few-shot prompting

Model	CROSSSUM-IN			XQUAD-IN		XORQA-IN-XX			XORQA-IN-EN		
	Higher	Medium	Low	Higher	Medium	Higher	Medium	Low	Higher	Medium	Low
<i>mT5</i> models – Fine-Tuned											
mT5-B	19.5	18.9	15.1	46.2	30.9	3.8	4.0	5.5	31.7	31.4	30.8
mT5-L	20.5	19.9	15.5	54.3	38.6	11.8	11.0	10.4	56.8	53.7	45.4
mT5-XL	22.7	21.1	15.3	57.4	40.5	20.7	13.5	15.6	58.2	56.2	46.5
mT5-XXL	25.9	24.2	10.4	62.0	44.4	28.8	23.6	21.9	70.3	68.9	59.1
<i>PaLM-2</i> models - Fine-Tuned											
PaLM-2-XXS	22.5	19.7	16.5	41.2	18.1	18.1	10.9	12.9	60.2	56.9	50.9
PaLM-2-XS	28.5	25.6	18.8	40.2	16.9	30.4	23.6	19.6	69.1	66.6	56.6
<i>PaLM-2</i> models - Few-shot prompted											
PaLM-2-XXS _{FS}	7.7	7.6	6.7	37.2	26.8	22.7	12.3	16.4	51.6	47.1	38.4
PaLM-2-XS _{FS}	18.4	16.4	13.0	65.1	53.3	39.2	32.0	29.5	67.0	65.3	56.5

Table 6: (Top) **Fine-tuning performance** of mT5 and PaLM-2 models (§3.5). **Bold** represents best numbers among fine-tuned models. PaLM-2 outperforms mT5 for longer-form generation task (CROSSSUM-IN), whereas mT5 models do well on short answer-span QA tasks. (Bottom) **Comparison of in-context learning vs. fine-tuning** on PaLM-2 models. In **Green**, we highlight the best PaLM-2 number (among fine-tuned and few-shot). For CROSSSUM-IN task requiring longer-form generation, fine-tuning outperforms few-shot prompting.

mT5 (older, but purpose-built for short-span generation) beats PaLM-2 on QA tasks even when fine-tuned; PaLM-2 wins on the longer-form CrossSum-IN. For QA, few-shot prompting on PaLM-2-XS already matches or beats fine-tuning it — for **long-form generation**, fine-tuning still wins outright, the one task where extra training data clearly pays off.

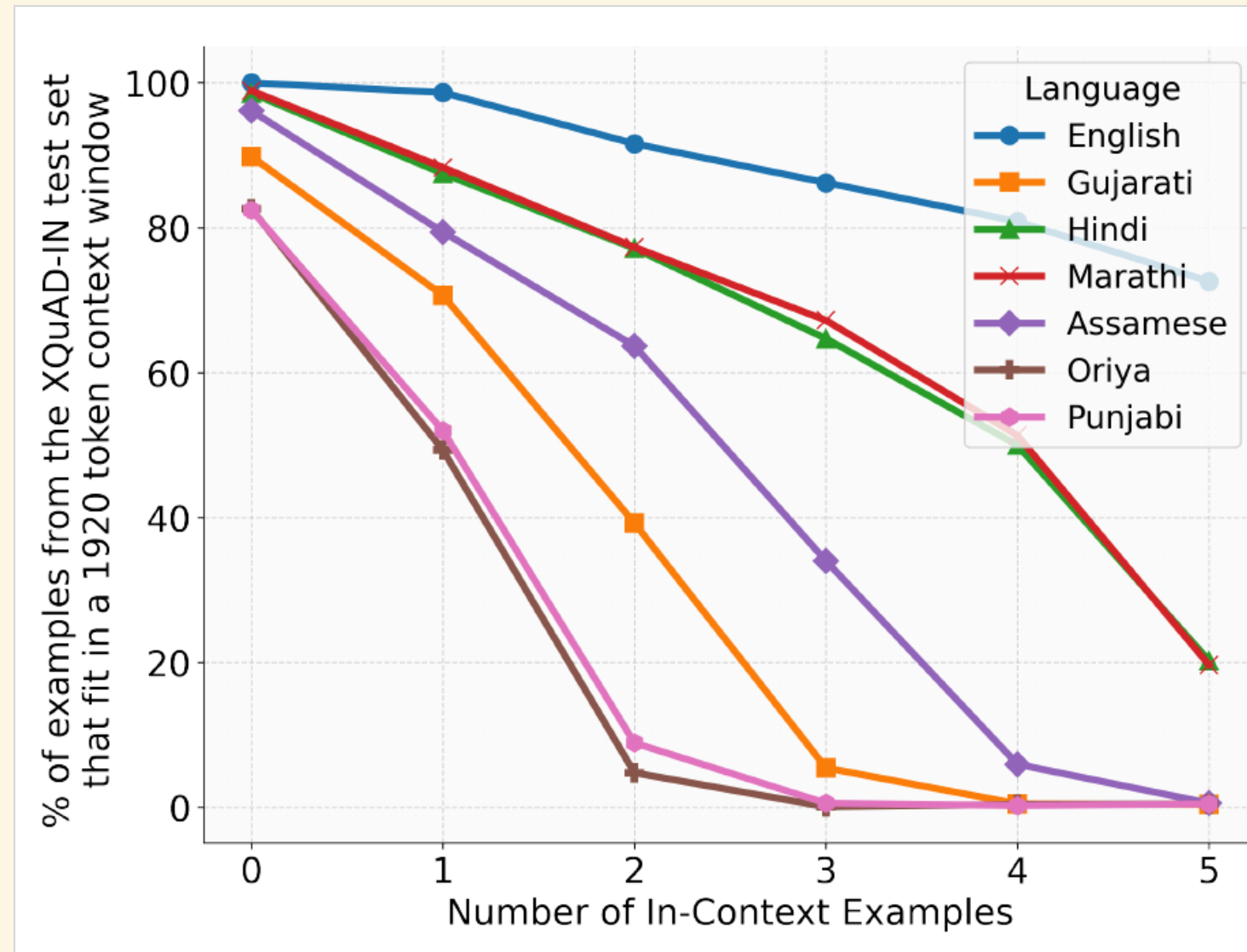
The tokenizer itself disadvantages some languages



Fertility = average number of sub-word tokens a word gets split into. English sits near 1; Pashto is lowest among Indic languages at 4.1; Tibetan is highest at **19.9** — a word in Tibetan costs roughly 20× the tokens of the same word in English.

Higher fertility means the same text eats more of the model's fixed context window — purely a tokenizer artifact, not a reflection of anything about the language itself.

What high fertility costs in practice



At a fixed 1920-token budget, English still fits 5 in-context examples for over 80% of its test set. Oriya and Punjabi — high-fertility languages — fall to **near 0%** by just 2–3 examples.

Combined with the previous slide, this directly explains the low-resource performance gap on the in-context-shots slide: it isn't only that models understand these languages less well — they also get to see fewer worked examples before answering.

What a single-language-tuned model buys you

there are only 160K Hindi Wikipedia articles. See Appendix §A for more details.

Language	CSSUM-IN			FLORES-IN (enxx)			XQUAD-IN			XORQA-IN-XX			XORQA-IN-EN		
	Airavata	LLaMA	BLOOMZ	Airavata	LLaMA	BLOOMZ	Airavata	LLaMA	BLOOMZ	Airavata	LLaMA	BLOOMZ	Airavata	LLaMA	BLOOMZ
en	31.6	19.1	13.9	-	-	-	75.7	49.2	86.8	69.5	56.0	72.2	69.5	59.7	68.7
hi	0.3	6.0	1.1	55.9	19.0	66.8	48.7	16.6	64.3	46.8	30.3	18.6	31.6	19.4	67.6
bn	0.0	2.0	0.2	8.5	9.6	73.9	6.5	2.5	57.9	0.2	3.8	0.3	33.8	12.6	70.0
pa	0.1	1.2	0.3	6.8	6.8	55.7	3.2	0.8	60.2	2.5	0.9	4.7	21.2	1.8	67.3
as	0.3	2.0	1.4	5.8	5.5	50.1	5.2	2.1	52.8	0.5	3.0	2.6	32.4	9.4	57.7
mni	0.2	2.3	0.2	6.1	6.7	6.5	-	-	-	1.4	2.9	0.3	22.6	5.4	21.5
sat	0.0	2.4	0.3	6.4	8.3	15.9	-	-	-	1.1	1.1	0.0	18.6	5.7	5.5

Airavata is fine-tuned almost exclusively on Hindi and English. It clearly beats LLaMA on Hindi — but its performance on every *other* Indic language (Bengali, Punjabi, Assamese, Manipuri, Santali) is much worse than BLOOMZ, a broad-coverage multilingual model. Specializing in one Indic language does not transfer to the rest.

Qualitative analysis: hallucination

The authors manually reviewed 20 PaLM-2-L predictions each, on CrossSum-IN and FLORES-IN, across five languages, with native speakers. In cross-lingual summarization, the model regularly invents details absent from the source article:

Summarize the following article in **Haryanvi**:

The White Garden, at Kensington Palace, was planted to mark 20 years since Princess Diana died in a car crash. The Duchess of Cambridge joined the princes on the garden tour. A spokeswoman for Kensington Palace said: "The engagement will allow the princes to pay tribute to the life and work of their mother." They met representatives from the causes and charities supported by Diana, including the Royal Marsden and Great Ormond Street hospitals, the National Aids Trust, Centrepont youth homelessness charity and the Leprosy Mission. Members of the public have been leaving tributes and flowers at the gates of the palace to mark the anniversary of Diana's death. The Princess of Wales died on 31 August 1997 in Paris, when William, now the Duke of Cambridge, was 15 and his brother was 12. The garden at their mother's former home has been inspired by memories of her life, style and image, such as her white "Elvis" Catherine Walker dress. The White Garden, as it is known, follows a tradition first established at Sissinghurst Castle in Kent, famous for its own white garden created in the 1930s. Their Royal Highnesses met gardener Sean Harkin who designed the display and Graham Dillamore who knew the princess when he worked there some 30 years ago. The garden has been open since spring and will continue into September with white roses, lilies, gladioli and cosmos. It is the fourth London memorial created in tribute to Diana - the others are the Diana Memorial Playground at Kensington Palace, the Diana Memorial Fountain in Hyde Park, and the Diana Memorial Walk at St James's Palace.

Model Output: प्रिंस विलियम और प्रिंस हैरी नै केंसिंगटन पैलेस मै **एक नया गार्डन खोलया** सै जो उनकी माँ, प्रिंसेस डायना की या
(English translation: Prince William and Prince Harry **have opened a new garden** at Kensington Palace in memory of their mother, Princess Diana.)

Qualitative analysis: bleeding into a related language

Awadhi, Haryanvi, and Chhattisgarhi are all related to Hindi and share its Devanagari script. Translating into them, the model frequently mixes in Hindi words and produces incorrectly inflected verb forms — errors highlighted in red below, against the human ground truth in blue:

Translate the following in **Awadhi**:

Input: One can only wonder what the keyboard will become when something newer comes along.

Model output: कोई केवल इ सोच सकत है कि जब कुछ नया आवत है त कीबोर्ड का होई।

Ground Truth: इ सिर्फ सोचि जाय सकत है कि जब कुछ नया आयी त कीबोर्ड का होई।

Translate the following in **Chhatisgarhi**:

Input: Scaffolds can be both virtual and real, in other words, a teacher is a form of scaffold but so is the little paperclip man in Microsoft Office.

Model output: मचान आभासी अउ वास्तविक दुनो हो सकत हंवय, दुसर शब्दमन में, एक शिक्षक मचान के एक रूप हवय लेकिन माइक्रोसॉफ्ट ऑफिस में छोटा पेपरक्लिप आदमी भी हवय।

Ground Truth: मचान आभासी अउ वास्तविक दूनों हो सकथे, दूसरे सब्द मन म, एकठन सिक्छ्क मचान के एक रूप हवय लेकिन माइक्रोसॉफ्ट ऑफिस म छोटा पेपरक्लिप आदमी घलो अइसा ही हवय।

Qualitative analysis: polysemy and missing information

In machine translation, the model sometimes commits to the wrong sense of an ambiguous English word, or simply drops information present in the source sentence:

Translate the following in **Konkani**:

Input: When asked for comment, Miller said, "Mike talks a lot during the **hearing**...I was getting ready so I wasn't really **hearing** what he was saying."

Model output: टिप्पणी करपाक विचारलें तेन्ना, मिलर म्हणाले, "माईक **आयकुपाच्या** वेळार खूब उलोवपाक लागता ... हांव तयार जातालो म्हणून तो कितें म्हणटालो हांव खरेंच **आयकनाशिल्लो**"

Ground Truth: टिप्पणी मागली तेन्ना मिलरान म्हणलें, "मायक **सुनावणी** वेळार खूब उलयता... हांव तयार जातालो म्हणून तो कितें सांगता तें म्हाका खरेंच **आयकूंक** नाशिल्लें"

Translate the following in **Assamese**:

Input: Casablanca is one of the least interesting places **to shop** in all of Morocco.

Model output: কেচাब্লাংকা হৈছে মৰক্কোৰ সকলো ঠাইৰ ভিতৰত আটাইতকৈ কম আকৰ্ষণীয় ঠাই।

Ground Truth: মৰক্কোত কাছাবলাংকা হৈছে **কিনা-কটা** কৰাৰ বাবে আটাইতকৈ আকৰ্ষণহীন ঠাই।

In the Konkani example, "hearing" means a court hearing the first time and "listening" the second — the model picks the "listening" sense for both. In the Assamese example, "to shop" is simply absent from the translation.

Dataset licensing

In compliance with the licenses of the original source datasets, IndicGenBench's evaluation data is released under three different licenses depending on lineage:

TASK	LICENSE	INHERITED FROM
XQuAD-IN, FLORES-IN	CC BY-SA 4.0	XQuAD, FLORES-200
CrossSum-IN	CC BY-NC-SA 4.0	CrossSum
XorQA-IN (both tasks)	MIT	XOR-TyDi

Building on existing benchmarks means inheriting their licensing constraints too — one more consequence of the translation-based approach from earlier in this deck.

Reading list

- [Singh, Gupta, Bharadwaj, Tewari & Talukdar, *IndicGenBench: A Multilingual Benchmark to Evaluate Generation Capabilities of LLMs on Indic Languages*](#) — ACL 2024, the source for every table and figure in this deck.
- [Bhattacharjee et al., *CrossSum*](#) — the cross-lingual summarization dataset CrossSum-IN extends.
- [NLLB-Team et al., *No Language Left Behind*](#) — source of FLORES-200, extended into FLORES-IN.
- [Artetxe, Ruder & Yogatama, *On the Cross-lingual Transferability of Monolingual Representations*](#) — introduces XQuAD, extended into XQuAD-IN.
- [Asai, Kasai, Clark, Lee, Choi & Hajishirzi, *XOR QA*](#) — source of XOR-TyDi, extended into XorQA-IN.
- [Doddapaneni et al., *IndicXTREME*](#) — the companion NLU/classification benchmark covered in the previous lecture.
- [Official IndicGenBench data and code.](#)